

Bishnu Prasad Pokharel

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SUMMARY

Automation and Controls Engineer with 1.5 years of hands-on experience independently designing, programming, commissioning, and validating industrial control systems in high-volume manufacturing environments. Experienced in full-lifecycle machine builds, PLC programming, HMI/SCADA development, electrical panel design, robotics integration, and vision system validation across cross-disciplinary engineering teams. Strong foundation in Structured Text, Ladder Logic, AutoCAD Electrical, SolidWorks, Creo, and industrial networking with a proven ability to self-learn new tools quickly and deliver production-ready systems.

EDUCATION

The University of Toledo, OH (Graduated: December 2025)

Grade Point Average: **3.9**

Bachelor of Science: Computer Science & Engineering

Honors & Awards: President's List (5 Semesters), Dean's List (All 9 Semesters), Ebeid Family STEM Scholar (3 times).

TECHNICAL SKILLS

Design and Electrical: AutoCAD Electrical (schematics, logic diagrams, I/O lists), SolidWorks (3D modeling), Creo (3D electrical cabinet design), Panel Design and Wiring, FMEA, Root Cause Analysis, DOE/Statistics

PLC and Controls: PLC Programming (Structured Text, Ladder Logic, MotionWorks IEC/Yaskawa; transferable to Rockwell Studio 5000/ControlLogix, Siemens S7), Servo Motors and Drives, VFDs, Motion Controllers, Level I/II Architecture, Hardware I/O Verification, Commissioning and Startup

HMI and SCADA: Movicon HMI/SCADA (real-time dashboards, failure analytics, operator interfaces), comparable to Wonderware/InTouch and FactoryTalk View SE, Ethernet/IP industrial networking, OPC UA (conceptual)

Vision and Robotics: Cognex Vision Systems (image processing, threshold calibration, quality validation), Keyence Safety Sensors and Light Curtains, Epson Robot coordination via Ethernet/IP, Laser Marker integration

Programming and Tools: Python, C/C++, Java, SQL, JavaScript, MATLAB, Git, Power BI, Minitab, Excel (Macros/VLOOKUP), Linux/UNIX

Relevant Coursework: Embedded Systems, Electric Circuits, Electronics I, Signals and Systems, Digital Logic Design, Computer Architecture, Data Communications, Operating Systems

EXPERIENCE

Tenneco Inc, Burlington, Iowa | Engineering Co-op

January 2023 – August 2024

- **Full-Lifecycle Machine Build:** Collaborated in a cross-disciplinary team of mechanical, computer, and electrical engineers to independently design and build a Spark Plug Torque Tester from scratch — owning component selection, electrical panel design, field wiring, servo motor and drive integration, Movicon HMI programming, and full plant commissioning. Delivered production-ready with 100% I/O verification and safety compliance.
- **Vision System Design and Validation:** Designed and validated Cognex Vision inspection protocols for spark plug print quality assurance, configuring image processing algorithms and establishing precise pass/fail thresholds that eliminated manual inspection bottlenecks and improved production accuracy.
- **Electrical Design and Documentation:** Self-taught AutoCAD Electrical from zero and produced the most complex and accurate schematics in the plant, including a \$1M+ program replacing 30 to 40 machines globally. Designed schematics for 3 machines from scratch and updated documentation for numerous existing systems during component replacements.
- **Robotics and Traceability Pipeline:** Integrated Epson robots and laser markers via Ethernet/IP into a real-time data acquisition and traceability pipeline with live Movicon HMI dashboards, achieving 100% batch traceability across the production line.
- **Safety Integration and Commissioning:** Led hardware commissioning and structured I/O verification for new and modernized equipment. Integrated Keyence safety sensors and light curtains with PLC control logic, ensuring 100% compliance with industrial safety standards.

UNIVERSITY PROJECTS

SilentBell Senior Design IoT Hardware System | MQTT, AutoCAD Electrical, SolidWorks, MERN Stack, 3D Printing

- **Full Hardware Lifecycle:** Designed electrical schematics in AutoCAD Electrical, selected all components, completed all wiring and assembly, modeled the enclosure in SolidWorks, and fabricated via 3D printing — same full-lifecycle build approach applied at Tenneco.
- **Control and Software Integration:** Architected a real-time MQTT pub/sub system with sub-second latency and greater than 99% event delivery reliability, validated through structured end-to-end testing.

MoneyMate Automated Data Pipeline | React Native, Node.js, MongoDB, OpenAI API, OCR

- Built a full-stack application integrating OpenAI API and OCR for automated data extraction at 95% accuracy, reducing manual entry by 80%.